

EC-444

An analogue bucket-brigade delay: warm, darkening repeats with drive in the loop.



Overview

EC-444 is a bucket-brigade (BBD) delay in the spirit of the budget analogue echo units of the late 70s and 80s. The input goes through a driven op-amp preamp, then into the delay line; every repeat passes back through the same drive and through the chip's clock filter, so the echoes get darker and grittier as they fade.

Two chips are on board: LONG gives delays from 50 to 550 ms, SHORT from 50 to 110 ms for slapback and doubling. As on a real bucket-brigade unit, turning TIME glides the delay rather than jumping, so the repeats bend in pitch while you turn it. Push REPEATS past 100% and the unit runs away into self-oscillation.

Reach for it for

- Warm, dark analogue echoes on guitars, vocals and synths
- Slapback on vocals and guitars (SHORT)
- Stereo doubling (DUAL CHIP)
- Dub-style runaway feedback and pitch bends
- Lo-fi, noisy repeats

Controls

PREAMP

TRIM	Input level, -24 to +12 dB.
DRIVE	The preamp's drive, 0 to 100% (30% by default). It saturates the input unevenly (the two halves of the wave clip differently, adding even harmonics) and also sits in the feedback loop, so each repeat is driven again.

DELAY

LONG / SHORT	The chip. LONG: 50 to 550 ms. SHORT: 50 to 110 ms, for slapback and doubling; TIME stays within that range.
TIME	The delay time. Changes glide smoothly, so the repeats bend in pitch as you turn it.
REPEATS	Feedback, 0 to 110% (40% by default). Above 100% the repeats grow instead of fading - self-oscillation.
DUAL CHIP	Runs the right side about 3% longer than the left, so the two sides' repeats fall slightly apart: a wider, doubled stereo echo.
OSC LIMIT	On by default. Holds the feedback in check so self-oscillation stays at a usable level instead of running away without limit.

TONE

CLOCK LPF	The low-pass filter on the repeats, 1 to 8 kHz (3.5 kHz by default). Because it is inside the loop, every repeat is darker than the one before. Lower it for murky, distant echoes; raise it for brighter ones.
NOISE	The chip's clock noise added to the repeats, 0 to 100% (15% by default).

MASTER

MIX	The dry / echo balance, 0 to 100% (35% by default). Up to 50% the dry signal stays at full level and the echo comes
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	up; from 50% the dry fades out, leaving only the echo at 100%.
DRIFT	See Drift below.
OUTPUT	The output level, -24 to +12 dB.
OUT meters	Output level, left and right. Click them to reset the peak hold.

The header

PRESETS	Opens the list of factory presets (or click the preset display). Choosing one sets everything except TRIM and OUTPUT.
◀ / ▶	Step through the presets.
Preset display	Shows the preset number and name. Click it for the list.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

EC-444 has no input guard and no output stage: the levels are set with TRIM and OUTPUT only.

Drift

DRIFT (0 - 100%, 50% by default) brings the unsteadiness of a real analogue unit: the chip's clock wanders, so the delay time breathes a little, and the clock filter - the tone of the repeats - slowly opens and closes. 50% is the standard amount, 100% twice that, 0 steady; the wander also speeds up as you turn it up.

Presets

Init	The default settings: LONG, 250 ms, 40% repeats.
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Slapback	SHORT, 95 ms, a single faint repeat, bright.
Dark Repeats	LONG, 380 ms, 60% repeats through a dark 1.8 kHz filter.
Dual Chip Doubler	SHORT, 60 ms, DUAL CHIP on: a wide doubled sound.
Ambient Wash	LONG, 520 ms, 78% repeats with DUAL CHIP and more drift.
Self-Oscillation	LONG, 300 ms, 105% repeats and more drive: the echoes build up on their own.
Lo-Fi Short	SHORT, 80 ms, heavy drive, a dark filter and lots of noise.
Clean Long	LONG, 450 ms, little drive, a bright 7 kHz filter and very little noise.

Tips

- For dub throws, set REPEATS around 90-105% and turn TIME by hand: the repeats swoop in pitch as the delay glides.
- If the echoes get too dark too fast, raise CLOCK LPF rather than REPEATS.
- On a send, set MIX to 100% so only the echo comes back.
- Use the SHORT chip with DUAL CHIP for a quick stereo double on a mono vocal or guitar.

File: gfx-Ec444.jsfx